Pico firmware:
– encoder, DSO/DS1, SD card change from shift register, SW4, SW5
– GPIO0: output 2MHz system clock
– GPIO3: output 100Hz irq

Connection block:
If shift register idea fails
reconnect Pico to encoder
(shift register works fine,
so this could be removed)

U3
Pico

J1
TCBM

U6
6510T

NOTE: DSO/DS1 connected through shift register
NOTE: P3 ACT LED goes between ACT and VCC

Pullups are necessary on all lines

U9
XC9572XL-VQ64

NOTE: ACK/DAV naming reversed to keep the TCBM2SD connector intact
NOTE: ACK here is controller's ACK output and goes to cable pin 13 (drive's DAV input)
NOTE: DAV here is controller's DAV input and goes to cable pin 11 (drive's ACK output)
NOTE: MODE==OE from GATE ARRAY

XR/W is OUTPUT: !((IR/W AND PHI2)
typical qualified R/W circuit for 6502 peripherals

U5
27C512

J1:
– switch chooses ROM
from 32K half of 27C512
closed = bottom half
open = top half

U4
KM62256CLP

Power section

J4
DC IN

J5
Module_Panel

ACT_3V3 – converted and inverted ACT
(the other side of LED is connected to GND)

* Resistor values need to be adjusted according to type
of LED installed on front panel. Sensible values for
bright LEDs is significantly higher than what is specified.

ACT is active low (LED goes to 5V)
ACT_3V3 is active high (LED goes to GND)

Q2
BSS138

J3
JTAG

Q1
BSS138

Q3
BSS138

Q4
BSS138

Q5
BSS138

Q6
BSS138

U19
74LS165

J6
SD CARD

23x18mm
SD Card module
3.3v

XS1
SD connector

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